- There are 7 Questions for 35 marks.
- Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class. You may use any result from class directly (without re-deriving the proof). Clearly justify any statement you make, even if the problem doesn't explicitly ask you to. Solving any multi-part question flawlessly will add 2 points to your total score for each such solution.
- Good luck!

1. [3] Consider the following statement:

   For every $\underline{x}, \underline{y} \in \mathbb{R}^n$, $\|\underline{x} - \underline{y}\|_p^2 + \|\underline{x} + \underline{y}\|_p^2 = 2\|\underline{x}\|_p^2 + 2\|\underline{y}\|_p^2$.

   For which values of $p \in \{1, 2, \infty\}$ is the above statement true?

2. [3] Suppose you are trying to estimate an unkown vector $\underline{x}$ given some structured linear measurements $\underline{y}$ of the form $\underline{y} = A\underline{x}$, for some (known, structured) matrix A. On searching through available algorithms in the literature, you find three algorithms, each of which comes with the below respective guarentees, for some fixed $\epsilon > 0$, and every $\underline{y} = A\underline{x}^\star$:

   - Algorithm 1 guarentees that the estimate $\hat{\underline{x}}$ provided by it satisfies $\|\underline{x}^\star - \hat{\underline{x}}\|_1 \leq \epsilon$
   - Algorithm 2 guarentees that the estimate $\hat{\underline{x}}$ provided by it satisfies $\|\underline{x}^\star - \hat{\underline{x}}\|_2 \leq \epsilon$
   - Algorithm 3 guarentees that the estimate $\hat{\underline{x}}$ provided by it satisfies $\|\underline{x}^\star - \hat{\underline{x}}\|_\infty \leq \epsilon$

   Assuming all three algorithms cost similar number of arithmetic operations, which algorithm will you use and why ?

3. [6] Given n non-zero (column) vectors $\underline{a}_1, \underline{a}_2, \ldots, \underline{a}_n$, the corresponding $n \times n$ distance matrix D is constructed such that the $(i,j)^{th}$ entry is given by $\|\underline{a}_i - \underline{a}_j\|_2^2$.
   (a) Express D as $D = D_1 + D_2 - D_3$, such that D_1 and D_2 have rank 1, and D_3 is a Gram matrix.
   (b) We are given the entries of the vectors $\underline{a}_1, \underline{a}_2, \ldots, \underline{a}_n$. For a given arbitrary vector $\underline{x} \in \mathbb{R}^n$, suppose we wish to compute $D\underline{x}$. How will you efficiently perform this computation? How many arithmetic operations will your procedure cost ? You can give your answer using the big Oh notation.

4. (a) [2] Given $m \times n$ matrices A, B and $\underline{y} \in \mathbb{R}^n$, what is the vector $\underline{x} \in \mathbb{R}^n$ that minimizes $\|A\underline{x} - B\underline{y}\|_2$?
   (b) [2] Suppose that any column of A is orthogonal to any column of B. How would you give a 'simple' equation (involving matrices A, B, transposes and matrix products) that captures this relationship ?
   (c) [2] In case A, B satisfy the property in (b) above, what is the corresponding answer to (a) ?

5. Consider the following sequence of matrices

$$A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A_n = \begin{pmatrix} A_{n-1} & I \\ I & -A_{n-1} \end{pmatrix} \text{ for } n \geq 2.$$

   (a) [1] What is the size of the matrix A_n ?
   (b) [2] Is A_n symmetric ?
   (c) [2] What is A_n^2 ? *Hint: Guess a pattern, then prove by induction*
   (d) [2] What is the rank of A_n ?

6. [4] A network consists of n links, labeled $1, \ldots, n$. A *path* through the network is a subset of the links (the order of the links on a path does not matter here).

   Each link has a (positive) *delay*, which is the time it takes to traverse it. We let $\underline{d}$ denote the n-vector that gives the link delays. The total travel time of a path is the sum of the delays of the links on the path. Our goal is to estimate the link delays (i.e., the vector $\underline{d}$), from a large number of (noisy) measurements of the travel times along different paths.

   This data is given to you as an $N \times n$ matrix P, where

$$P_{ij} = \begin{cases} 1, & \text{if link } j \text{ is on path } i, \\ 0, & \text{otherwise,} \end{cases}$$

   and an N-vector $\underline{t}$ whose entries are the (noisy) travel times along the N paths. You can assume that $N > n$.

   You will choose your estimate $\hat{\underline{d}}$ by minimizing the Euclidean norm between the measured travel times ($\underline{t}$) and the travel times predicted by the sum of the link delays.

   Explain how to do this, and give a matrix expression for $\hat{\underline{d}}$. If your expression requires assumptions about the data P or $\underline{t}$, state them explicitly.

7. (a) [3] Recall that the accumulated amount with a compound interest r and Principal P after t years is given by

$$P\left(1 + \frac{r}{n}\right)^{tn},$$

   where n is the compounding frequency ($n = 1$ if the compounding is done annually, $n = 12$ the compounding is done monthly, etc). Suppose the compounding is done continuously (i.e. $n \to \infty$), obtain the expression for the accumulated amount in terms of the principal P, annual interest rate r and time t in years.

   (b) [3] Consider a $2n$ dimension box with each side of width 2, centered at the origin. Let B_n be the largest possible ball (centered at the origin) that can placed inside the box. As $n \to \infty$, what fraction of the $2n-$ dimensional volume of the box is occupied by the ball B_n?

   You may use the following: The $2n-$dimensional volume of a ball of radius 1 is $\pi^n/n!$, and the $2n-$dimensional volume of a box with each side 2 is 2^{2n}.